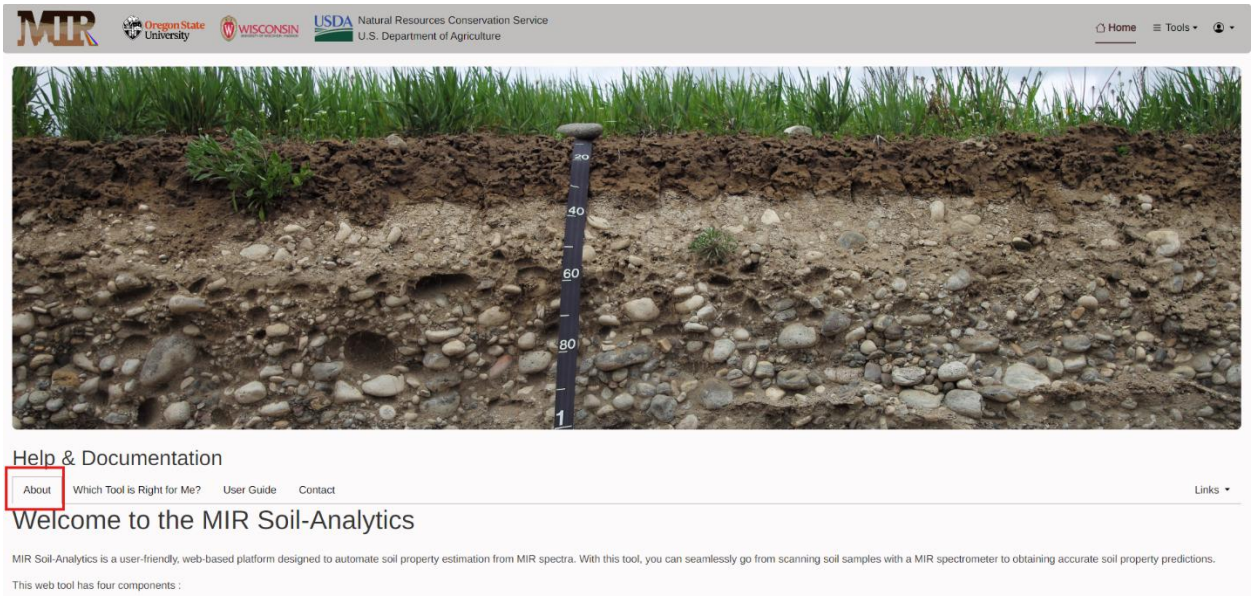
# MIR Soil Prediction Platform – User Guide

## Home page – About




### Welcome to the MIR Soil Prediction Platform

The MIR Soil Property Platform is a user-friendly, fast, web-based platform for soil property estimation from MIR spectra. After uploading MIR spectra, some preprocessing, and choosing and executing a prediction model, this platform will provide soil property estimates with a prediction accuracy.

This platform has four components

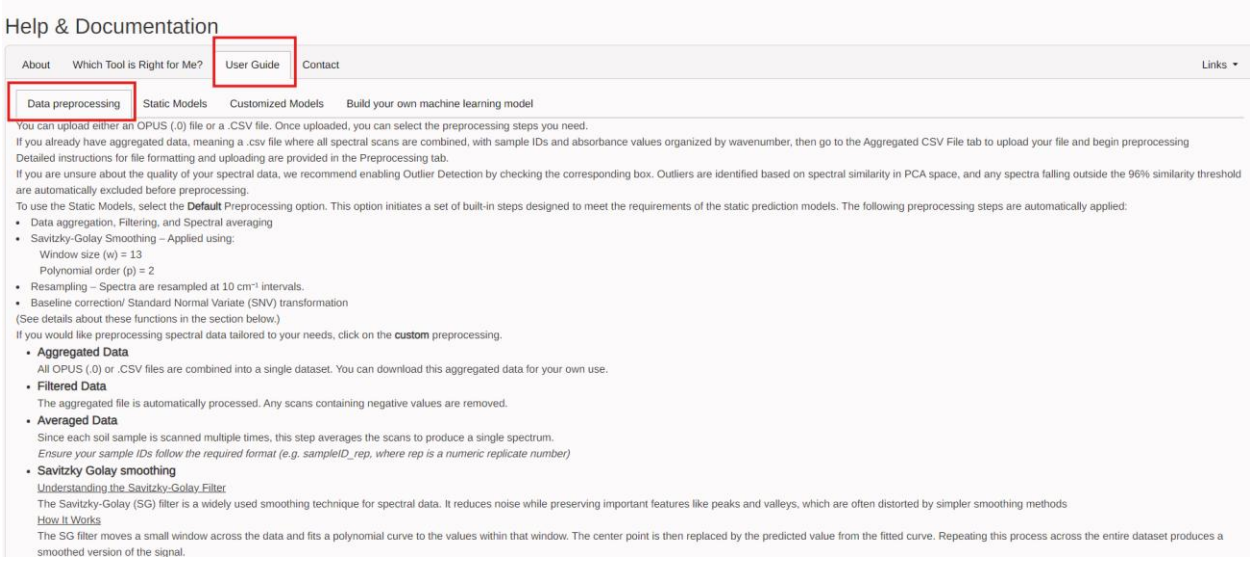
- **Data Preprocessing** - Prepare and preprocess your spectral data for modeling
- **Static models** - Make soil property predictions using built-in models, developed using the USDA NRCS NSSC-KSSL mid-infrared (MIR) spectral library.
- **Customized models** -Generate tailored predictions from your spectral dataset using the K-nearest neighbor algorithm to identify and subset spectrally similar samples from the NSSC-KSSL spectral library.
- **Build your machine learning models** - Develop and train custom models tailored to your dataset for advanced soil analytics.

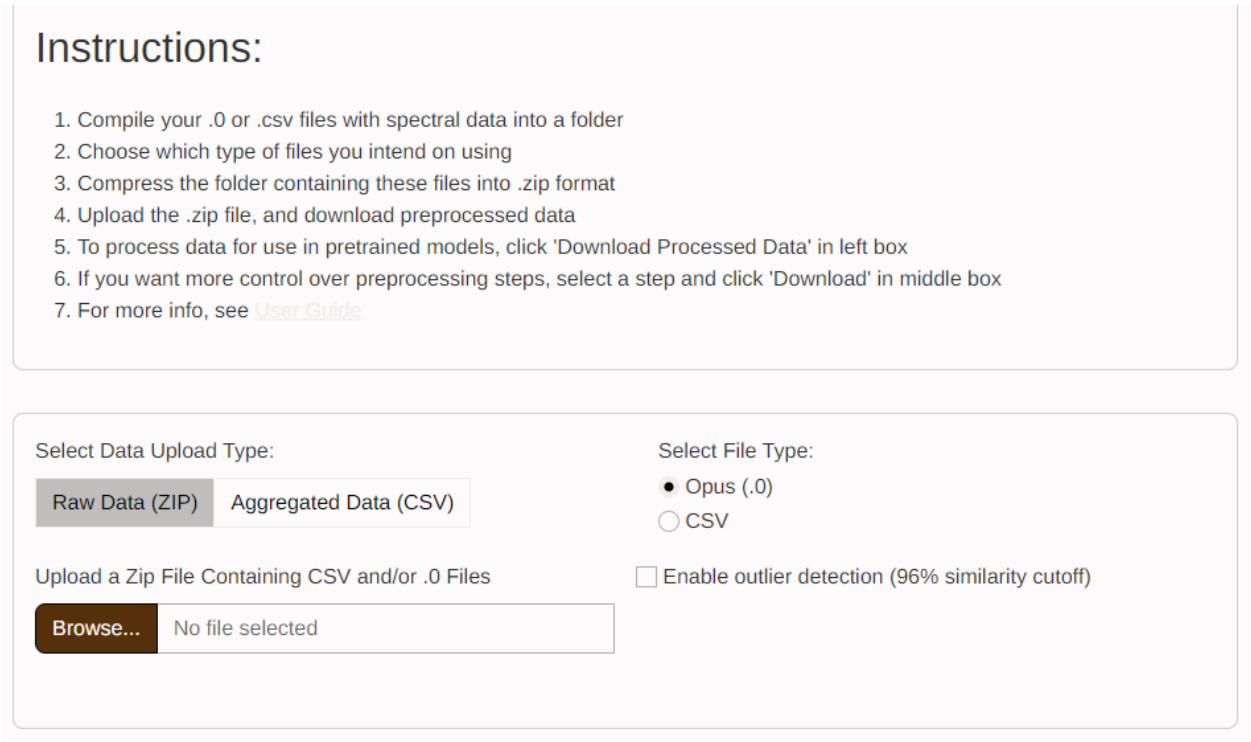
To access each component, use the dropdown menu under the **Tools** tab in the top-right corner of the page.

Let MIR Soil Prediction platform handle the modelling so that you can focus on results.

Need help choosing the right tool for your dataset? Use our decision tree to navigate the web tool efficiently. Visit **Which Tool is Right for Me?** to get started!

## User Guide - **Date Preprocessing**



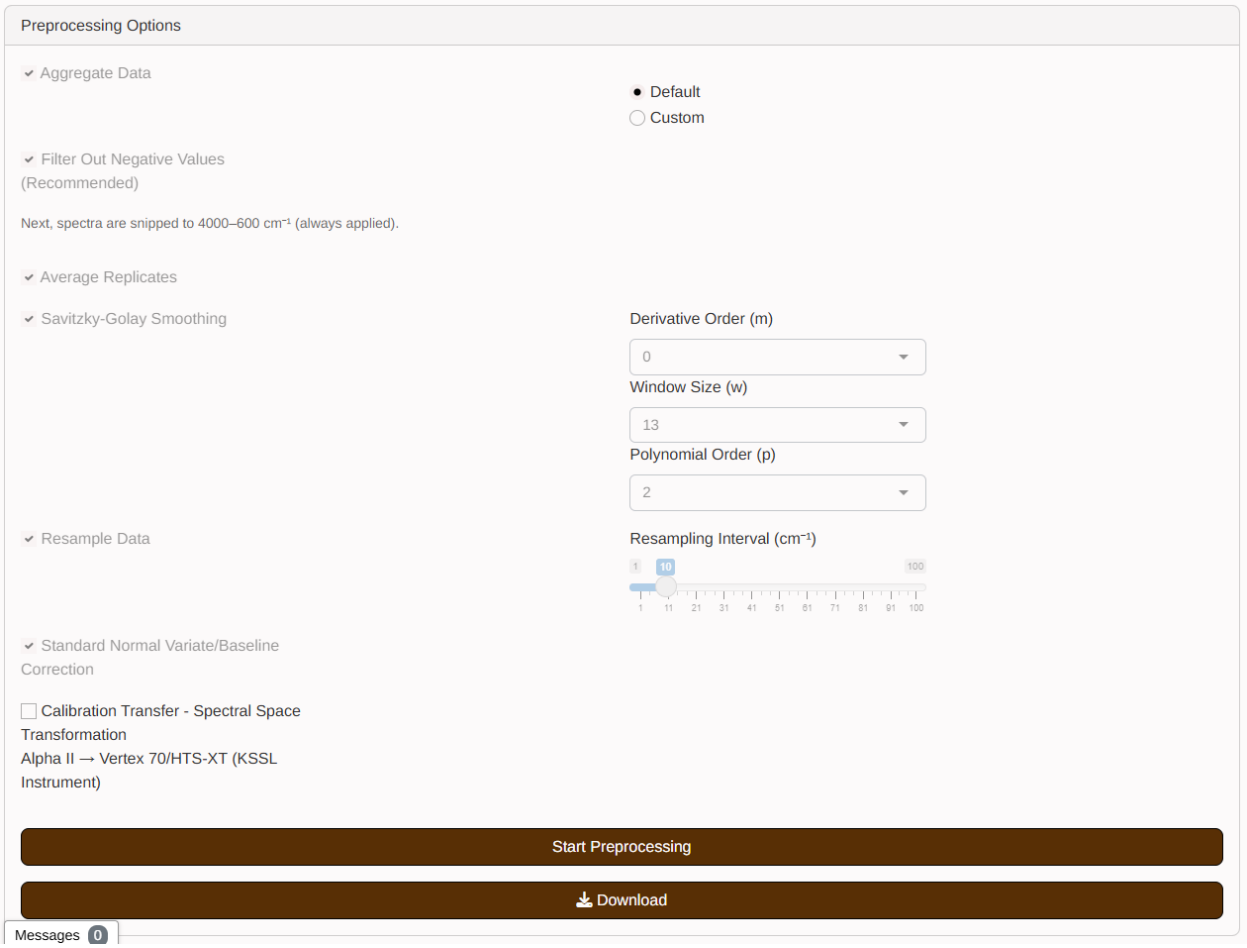




You can upload either an OPUS (.0) file or a .CSV file.

If you already have aggregated .csv file containing all spectral scans, including sample IDs and absorbance values organized by wavenumber, then go to the Aggregated CSV File tab to upload your file and begin preprocessing

Detailed instructions for file formatting and uploading are provided in the Preprocessing tab.

If you are unsure about the quality of your spectral data, we recommend enabling Outlier Detection by checking the corresponding box. Outliers are identified based on spectral similarity in PCA space, and any spectra falling outside the 96% similarity threshold are automatically excluded before preprocessing.




To use the Static Models, or customize them using the Customized Model component, select the **Default Preprocessing option.**

This option initiates a set of built-in steps designed to meet the requirements of the static prediction models. The following preprocessing steps are automatically applied:

* Data aggregation, Filtering, and Spectral averaging

* Savitzky-Golay Smoothing – Applied using:

    Window size (w) = 13

    Polynomial order (p) = 2

* Resampling – Spectra are resampled at 10 $cm^{-1}$ intervals.

* Baseline correction/ Standard Normal Variate (SNV) transformation

(See details about these functions in the section below.)

* Optional: Calibration Transfer – Spectral Space Transformation (SST)

Alpha II benchtop spectrometer with a front-reflectance module (Bruker Optics) → Vertex 70/HTS-XT (Bruker Optics / KSSL instrument)

As explained in the Static Models section, the prediction models were developed using the KSSL spectral library, with spectra collected using a Bruker Vertex 70/HTS-XT instrument.

If your spectra were collected using a different instrument, particularly a Bruker Alpha II, instrumental differences may affect prediction accuracy. In such cases, you can select the Calibration Transfer (SST) option to transform your spectra to more closely resemble those collected using the KSSL instrument.

For more information about the preprocessing requirements, performance, and limitations of SST, refer to the Calibration Transfer – Spectral Space Transformation (SST) section of this user guide or the corresponding section on the website.

If you would like preprocessing spectral data tailored to your needs, click on the **Custom Preprocessing** option.

- **Aggregated Data**
  All OPUS (.0) or .CSV files in the zip file are combined into a single dataset. You can download this aggregated data for your own use.

- **Filtered Data**
  The aggregated file is automatically processed. Any scans containing negative values are removed.

- **Averaged Data**
  Since each soil sample is scanned multiple times for quality control, this step averages the scans to produce a single spectrum. Once this is applied the spectra is also snipped from 4000 – 600 cm-1 which considered as the MIR region.
  *Ensure your sample IDs follow the required format (e.g. sampleID_rep, where rep is a numeric replicate number)*

- **Savitzky Golay smoothing**

  Understanding the Savitzky-Golay Filter

  The Savitzky-Golay (SG) filter is a widely used smoothing technique for spectral data. It reduces noise while preserving important features like peaks and valleys, which are often distorted by simpler smoothing methods

How It Works

The SG filter moves a small window across the data and fits a polynomial curve to the values within that window. The center point is then replaced by the predicted value from the fitted curve. Repeating this process across the dataset produces a smoothed version of the signal.

The filter depends on three parameters:

| Parameter | Meaning |
|---|---|
| Window size (w) | Number of data points used in each local fit (must be an odd number). A smaller window retains fine detail but smooths less. A larger window provides stronger smoothing but may flatten sharp features. |
| Polynomial degree (p) | Degree of the polynomial fitted within each window. It should be lower than w and typically 2 or 3. |
| Derivative order (m) | Order of the derivative to compute.<br>Use m = 0 for smoothing only.<br>m = 1 for first derivative, and m = 2 for second derivative. |

Best Practices

- Start with the defaults and check if the result looks clean but not over-smoothed.
- Avoid using a high-degree polynomial with a small window, as this can overfit noise.
- Use visual inspection to see whether fine-tuning is needed for modeling or interpretation.

The SG filter is an essential tool for improving signal quality while preserving meaningful structures of the spectral data. By choosing the parameters, you can improve the data for further analysis and modeling.

- **Resampling**

Why Resample MIR Spectra?

MIR spectra are often recorded at very high resolution, capturing absorbance values at hundreds or thousands of wavenumbers. While this level of detail can be valuable, it also introduces noise, redundant information, and large file sizes. These could complicate data processing and slow down modeling workflows. Resampling addresses this by reducing

the number of data points, and it selects absorbance values at regular intervals (e.g., every 10 $cm^{-1}$), while preserving the essential shape and features of the spectrum.

Keep in mind:

If you resample at large intervals, you lose important spectral features that are critical for modeling or interpretation.

- **Standard Normal Variate (SNV) transformation**

  Standard Normal Variate (SNV) / Baseline Correction is a normalization technique used to correct unwanted variation in spectral data. SNV transforms each spectrum so that it has a mean of 0 and a standard deviation of 1, making spectra more consistent and comparable.

  Why Use SNV?

  When MIR spectra are collected, soil samples can scatter light differently because of surface texture, particle size, or packing. This often results in vertical shifts (offsets in absorbance) and slope differences across spectra. These variations are unrelated to the actual chemical composition, and that is why SNV becomes useful.

  SNV helps to reduce scatter effects, remove baseline offsets and slope variations, and make the spectra more comparable and interpretable.

  Improving signal clarity and consistency, SNV/baseline correction enhances the reliability of downstream analysis, and it is recommended preprocessing the MIR data up to the baseline correction step.

**How to Use Calibration transfer - Spectral Space Transformation (SST)**

**When should you use SST?**

The prediction models (Static models) available on this platform were developed using the USDA–NRCS Kellogg Soil Survey Laboratory (KSSL) spectral library, with spectra collected using a Bruker Vertex 70/HTS-XT instrument (Bruker Optics).

If your spectra were collected using a different instrument, differences in instrumental characteristics may introduce spectral variations and affect prediction accuracy.

To minimize these differences, the platform provides a **Spectral Space Transformation (SST)** option. SST transforms spectra collected using the Alpha II (Bruker Optics) instrument to more closely resemble those collected using the KSSL Vertex 70/HTS-XT instrument, improving spectral compatibility with the existing prediction models.

**Apply the required preprocessing.**

Before using SST, your spectra needs to be preprocessed using the same sequence as the platform's default preprocessing workflow. Or if you'd like to customize it, once your individual or aggregated spectral files are uploaded, make sure of doing

1. **Spectral range selection:** 4000–600 $cm^{-1}$ (automatically selected by the application)
2. **Savitzky–Golay (SG) smoothing:** Window size = 13, polynomial order = 2
3. **Spectral resampling:** 10 $cm^{-1}$ intervals
4. **Standard Normal Variate (SNV) normalization**

These four preprocessing steps are **required** for SST compatibility. Filtering negative absorbance values and averaging replicate spectra are optional and are not mandatory for applying SST.

After completing the required preprocessing, select the **Spectral Space Transformation (SST)** option to transform your spectra before applying the KSSL prediction models.

**How effective is SST?**

In our evaluation using paired spectra from the Alpha II spectrometer and Vertex 70/HTS-XT instruments, SST improved spectral agreement between the two instruments:

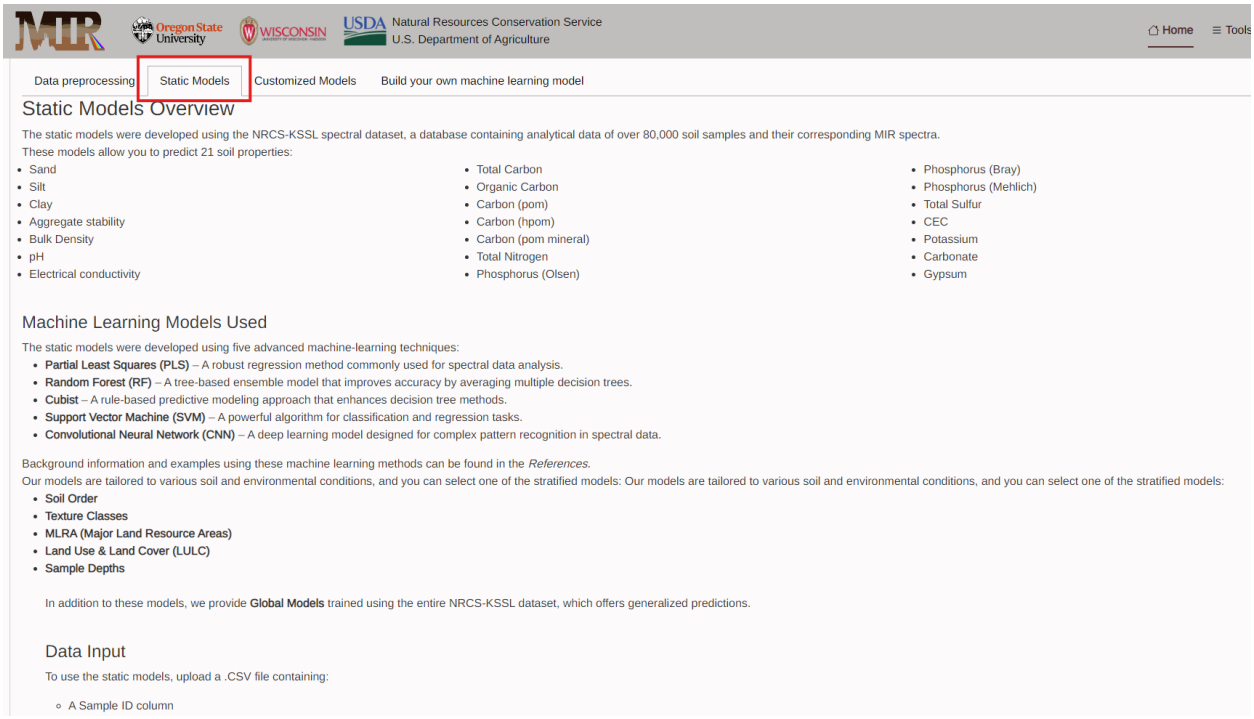
- **Spectral RMSE decreased from 0.280 to 0.212**, representing approximately a 24% reduction in spectral differences.
- **Concordance Correlation Coefficient (CCC) increased from 0.82 to 0.88**, indicating improved spectral agreement.

These results demonstrate that SST can reduce instrument-related spectral differences, potentially improving the transferability of KSSL prediction models to Alpha II spectra.

**Important note: Instrument compatibility**

*The SST transformation was developed using 100 soil samples scanned on one Bruker Alpha II and one Bruker Vertex 70/HTS-XT instrument. It has not yet been independently validated on other Alpha II units or other instrument models. Because individual instruments may have different optical characteristics, transformation performance may vary. Users are encouraged to evaluate prediction accuracy using reference soil samples when applying SST to spectra from a different instrument.*

# User Guide – Static models




## Static Models Overview

The static models were developed using the NRCS-KSSL spectral dataset, a database containing analytical data of over 80,000 soil samples and their corresponding MIR spectra. These models allow you to predict 19 soil properties:

- Sand
- Silt
- Clay
- Aggregate stability
- Bulk Density
- pH
- Electrical conductivity
- Total Carbon
- Estimated Organic Carbon
- Carbon (hpom)
- Carbon(maom)
- Total Nitrogen
- Phosphorus (Olsen)
- Phosphorus (Bray)
- Phosphorus (Mehlich)

- Total Sulfur
- CEC
- Potassium
- Carbonate
- Gypsum

## Machine Learning Models

Static models were developed using five advanced machine-learning techniques.

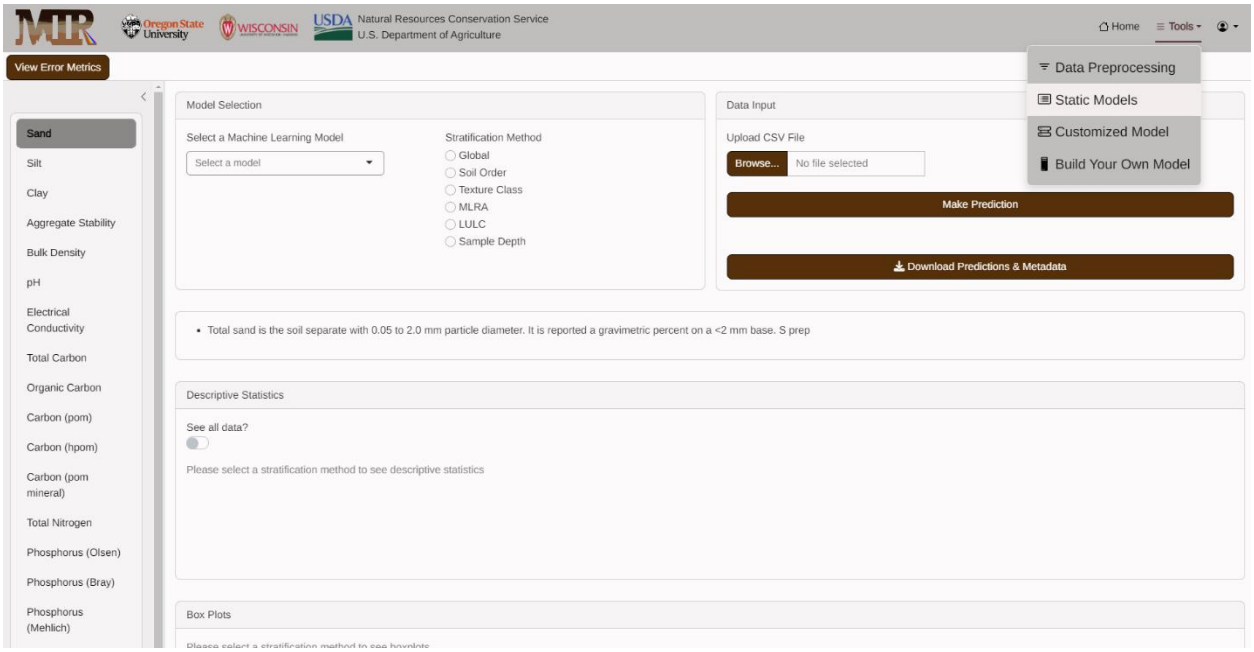
- **Partial Least Squares (PLS)** – A robust regression method commonly used for spectral data analysis.
- **Random Forest (RF)** – A tree-based ensemble model that improves accuracy by averaging multiple decision trees.
- **Cubist** – A rule-based predictive modeling approach that enhances decision tree methods.
- **Support Vector Machine (SVM)** – A powerful algorithm for classification and regression tasks.
- **Convolutional Neural Network (CNN)** – A deep learning model designed for complex pattern recognition in spectral data.

Background information and examples using these machine learning methods can be found in the *References*

Our models are tailored to various soil and environmental conditions, and you can select one of the stratified models:

- **Soil Order**
- **Texture Classes**
- **MLRA (Major Land Resource Area)**
- **Land Use & Land Cover (LULC)**
- **Sampling Depths**.

In addition to these models, we provide **Global Models** trained using the entire NRCS-KSSL dataset, which offers generalized predictions.




## Data Input

To use the static models, upload a .CSV file containing:

- A Sample ID column
- Preprocessed spectral data resampled at every 10 $cm^{-1}$

Below is an example of the required CSV format:

| **Column** | **Description** |
|---|---|
| Sample ID | Unique identifier for each soil sample |
| 4000 | Absorbance at 4000 $cm^{-1}$ |
| 3980 | Absorbance at 3990 $cm^{-1}$ |
| … | … |
| 600 | Absorbance at 600 $cm^{-1}$ |

If your spectral data is not preprocessed, use the **Data Preprocessing** tab in the web tool to generate a compatible preprocessed .CSV file.

If your spectral data is not yet preprocessed follow these steps:

- Use the **Data Preprocessing** tab – The Default preprocessing option in the web tool to generate a compatible preprocessed .CSV file.
- Compress your OPUS (.0) or .CSV files into a .zip file and upload it in the Data Preprocessing tab
- Choose the “Default” button in the custom preprocessing options.
- Click "Download" (located at the bottom left) and use the downloaded file for predictions.

## Generate Soil Property Predictions

From the left sidebar, choose the soil property you want to predict. From the Machine Learning drop-down menu, select your preferred model.

**Choose a Model Type** – Use the radio buttons to select:
• Global Models (trained on all available data)
• Stratified Models (Soil Order, Texture Class, MLRA, LULC, Sample Depth)

If a stratified method is chosen, select the appropriate stratified model from the drop-down menu based on the stratification. If a specific stratification model is not available, it means the stratified dataset did not have enough data to develop a machine learning model for that category.

## View and Download Predictions

Predictions will be displayed at the bottom of the page. Click the Download button to save the predictions and their metadata.

## How to Choose the Right Model

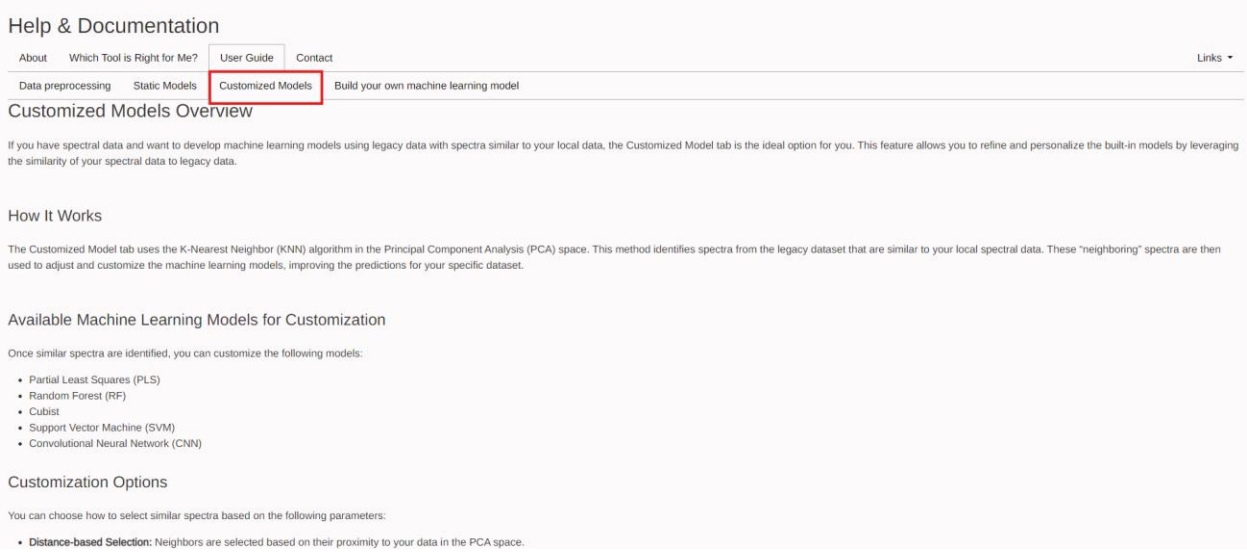
To help you select the most suitable model, we offer several tools for evaluating model performance and data compatibility:

- **Review Model Performance**
  *Error Matrices*: Located at the top left of the page, these matrices display the validation results using an independent dataset to help you assess model accuracy. The static models were trained using 70% of the data and tested on the remaining 30%. The accuracy matrix reflects the model's performance on the validation set.
- **Check Spectral Compatibility**
  *Spectral Plots*: These plots overlay your spectral data with the spectral data from the selected model, allowing visual comparison. If the spectra do not align well, consider

applying the Spectral Space Transformation (SST) option available in the Preprocessing component to reduce potential instrument-related spectral differences and improve spectral compatibility. This option is primarily intended for spectra collected using a Bruker Alpha II instrument.
*PCA Plots*: Principal Component Analysis (PCA) Plots compare your data with the dataset used to train the model. If your data points are far from the KSSL data in the PCA plot, consider selecting a different model for better predictions.

- **Explore Dataset Descriptive Statistics**
  Navigate to the Descriptive Statistics tab to view summary statistics of the dataset used to build the model. By default, this tab displays statistics for the selected model. To compare multiple models, click <See All Data>.
- **Check Spatial Information**
  The map shows the location of the training data. Clicking on any point displays detailed information, including taxonomic data.

## User Guide – Customized models




### Customized Models Overview

The Customized Model tab allows you to build machine learning models using KSSL data with spectra similar to your local data. It refines and personalizes the built-in models by leveraging the similarity in your spectral data and the KSSL spectral library.

How It Works

The Customized Model tab uses the K-Nearest Neighbor (KNN) algorithm in the Principal Component Analysis (PCA) space. It identifies spectra from the KSSL dataset similar to your local spectral data. The neighboring spectra are then used to adjust and customize the machine learning models, improving the predictions.

## Available Machine Learning Models for Customization

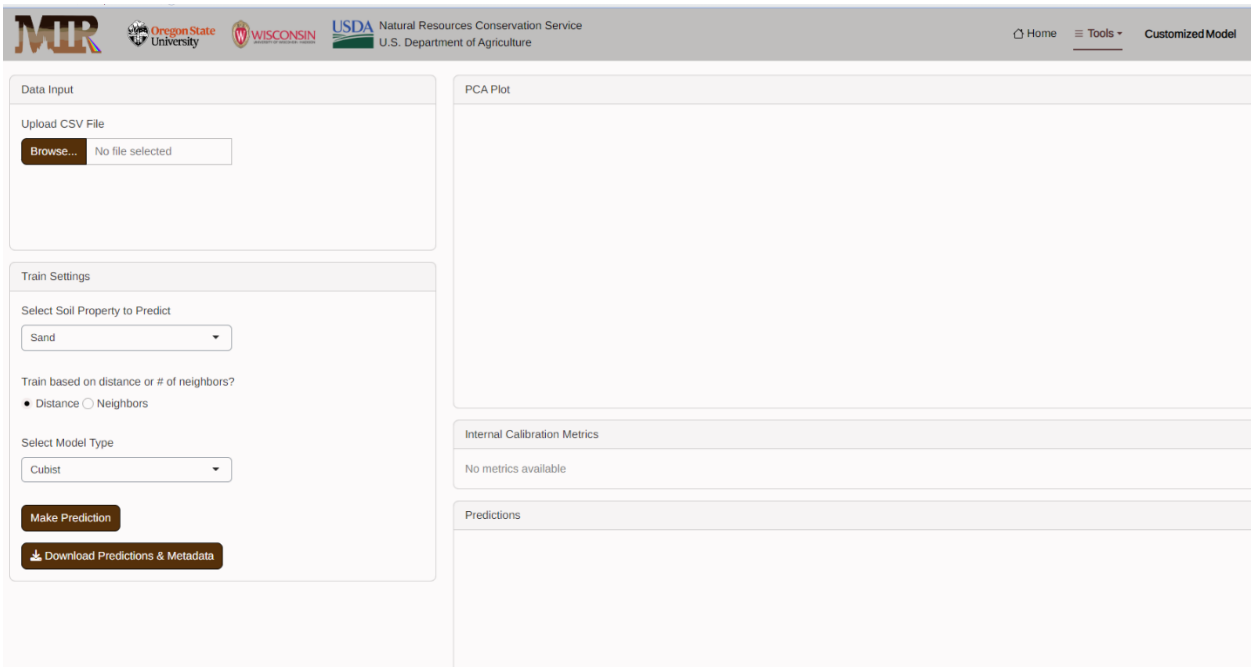
Once similar spectra are identified, you can customize the following models:

- Partial Least Squares (PLS)
- Random Forest (RF)
- Cubist
- Support Vector Machine (SVM)
- Convolutional Neural Network (CNN)

You can choose how to select similar spectra based on the following parameters:

- **Distance-based Selection:** Neighbors are selected based on their proximity to your data in the PCA space.
- **Number of Neighbors:** You can specify the exact number of similar spectra (neighbors) to use in the customization process.

You can fine-tune the model to better suit your local data and improve the prediction accuracy. For datasets with few samples, we recommend using the Distance-based Selection, but the platform is unable to provide validation metrics for this feature. You can check the model's internal calibration metric in the Calibration metric tab.




## Data Input

The data input process is identical to that of the static models. Upload a .CSV file containing:

- A Sample ID column
- Preprocessed spectral data resampled at every 10 $cm^{-1}$

Below is an example of the required CSV format:

| **Column** | **Description** |
|---|---|
| Sample ID | Unique identifier for each sample |
| 4000 | Absorbance at 4000 $cm^{-1}$ |
| 3980 | Absorbance at 3990 $cm^{-1}$ |
| … | … |
| 600 | Absorbance at 600 $cm^{-1}$ |

If your spectral data is not yet preprocessed follow these steps:

- Use the **Data Preprocessing** tab – Default preprocessing step in the web tool to generate a compatible preprocessed .CSV file.
- Compress your OPUS (.0) or .CSV files into a .zip file
- Upload it in the Data Preprocessing tab.
- Click "Download Processed Data" (located just below the data upload button on the left), and use the downloaded file for predictions.

### Generate Your Soil Property Predictions

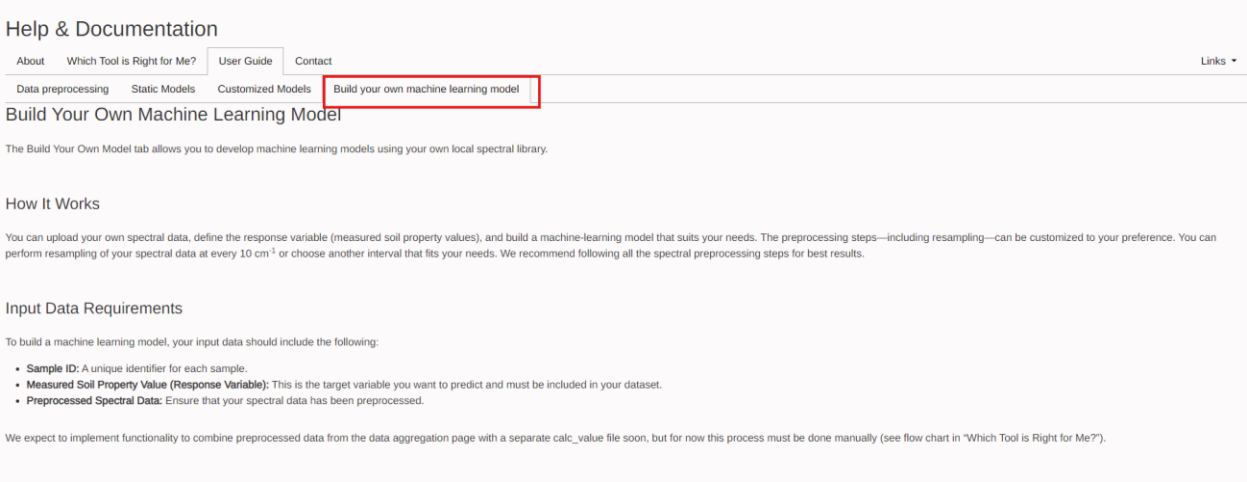
In the Train settings panel (bottom left, just below Data Input), start by selecting the soil property you're interested in from the dropdown menu. Then, configure the neighbor selection by using the radio buttons and slider to choose either distance-based or choosing a fixed number of neighbors per PCA point.

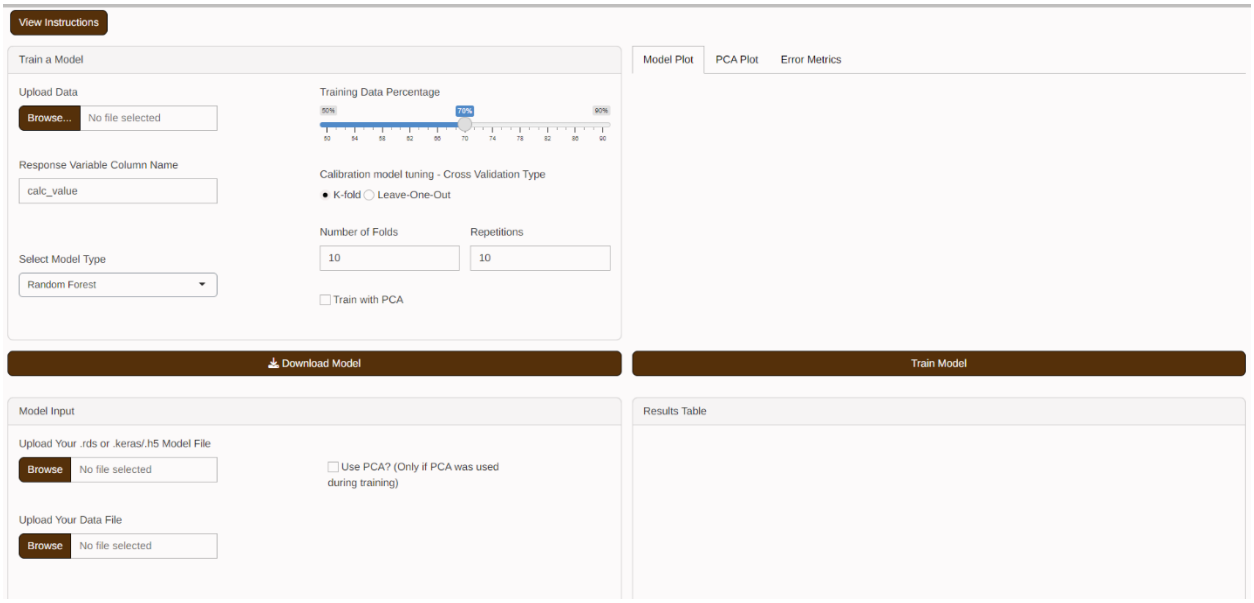
The PCA plot displays the KSSL data, the input data you upload (overlaid on top of the KSSL data points), and the subset of KSSL data points selected by the algorithm based on your neighbor selection. The more neighbors you select, the longer it will take for the model to train and predict.

Once the neighbors are selected, choose a Model Type using the corresponding dropdown to specify which machine learning model you'd like to train. Then click make prediction.

Once you click “Make Prediction”, the model will begin training and once finished, will display predictions in the bottom left. You may download the predictions by clicking “Download Predictions”.

## User guide - Build Your Machine Learning Model






The Build Your Model tab allows you to develop machine learning models using your local spectral library.

How it works

You can upload your spectral data, specify the response variable (e.g., measured soil property values), and build a machine learning model tailored to your needs. You also have the option to preprocess your spectral data using customizable settings. Visit the Data Preprocessing tab to get started.

## Input Data Requirements

To build a machine learning model, your input data should include the following:

- **Sample ID:** A unique identifier for each sample.
- **Measured Soil Property Value (Response Variable):** This is the target variable you want to predict and must be included in your dataset.
- **Preprocessed Spectral Data:** Ensure that your spectral data has been preprocessed.

## Training the Model

The Build Your Model tab allows you to train machine learning models using your spectral data and measured soil property values. Follow the steps below to build and optimize your model:

1. **Upload Your Data**
   - Ensure that your spectral data is preprocessed before uploading.
   - Upload a dataset containing both spectral data and measured soil property values (response variable).

2. **Specify the Response Variable**
   - Identify the response variable (the soil property you want to predict) by entering its column name into the web tool.

3. **Choose the data splitting method for calibration and validation.**

   First, use the slider to select the percentage of data to be used for training the model. Then, choose the cross-validation method for internal model tuning. You can choose between K-fold cross-validation and leave-one-out cross-validation. We recommend using K-fold cross-validation with 10 folds and 10 repetitions for a good balance between computational efficiency and accuracy. The best model is selected based on the lowest RMSE (Root Mean Square Error).

4. **Select a Machine Learning Algorithm**
   - Choose a model from the Select Model Type dropdown menu:
     - Partial Least Squares (PLS)
     - Random Forest (RF)
     - Cubist
     - Support Vector Machine (SVM)
     - Convolutional Neural Network (CNN)

5. **Dimensionality Reduction (Optional)**
   - If you choose RF, SVM, or Cubist, you may reduce data dimensions using PCA (Principal Component Analysis).
   - Why Use PCA? It enhances training speed and efficiency.
   - When NOT to Use PCA: PCA is not recommended for CNN and PLS models, as it negatively impacts their performance.
   - How to Apply PCA: If using RF, SVM, or Cubist, check the "Train with PCA" box before training.

6. **Train the Model**
   - Click Train Model to start the training process.
   - A progress bar will indicate the training status.

7. **Review Model Performance**
   Once training is complete, you can analyze:
     - Model Plot: A visual representation of training performance.
     - PCA Plot: Shows how your data aligns with the trained model.
     - Calibration/Validation Error Metrics: Assess model accuracy.

8. **Download the Model**
   - Click Download Model to save your trained model.
   - If you trained with PCA, a ZIP file will be downloaded containing both the trained model and the PCA transformation file.

## Using the Model

Below the model training section, there is a box that allows you to upload a .rds, or .keras/.h5 model (for CNN). If PCA was used during training, click the Use PCA box and upload the .rds PCA model as well. Finally, upload your data file. This should be in a similar format to the files uploaded in the static models. ensure your data file matches the expected format of your model (for example, if your model was trained using data resampled at 50 $cm^{-1}$, ensure your spectral data matches that). Also, ensure you have a Sample ID column at the beginning.

Once this is done, click “Run Custom Model”, and your predictions will appear in the results table to the right. You may download the predictions by clicking the “Download Predictions” button.